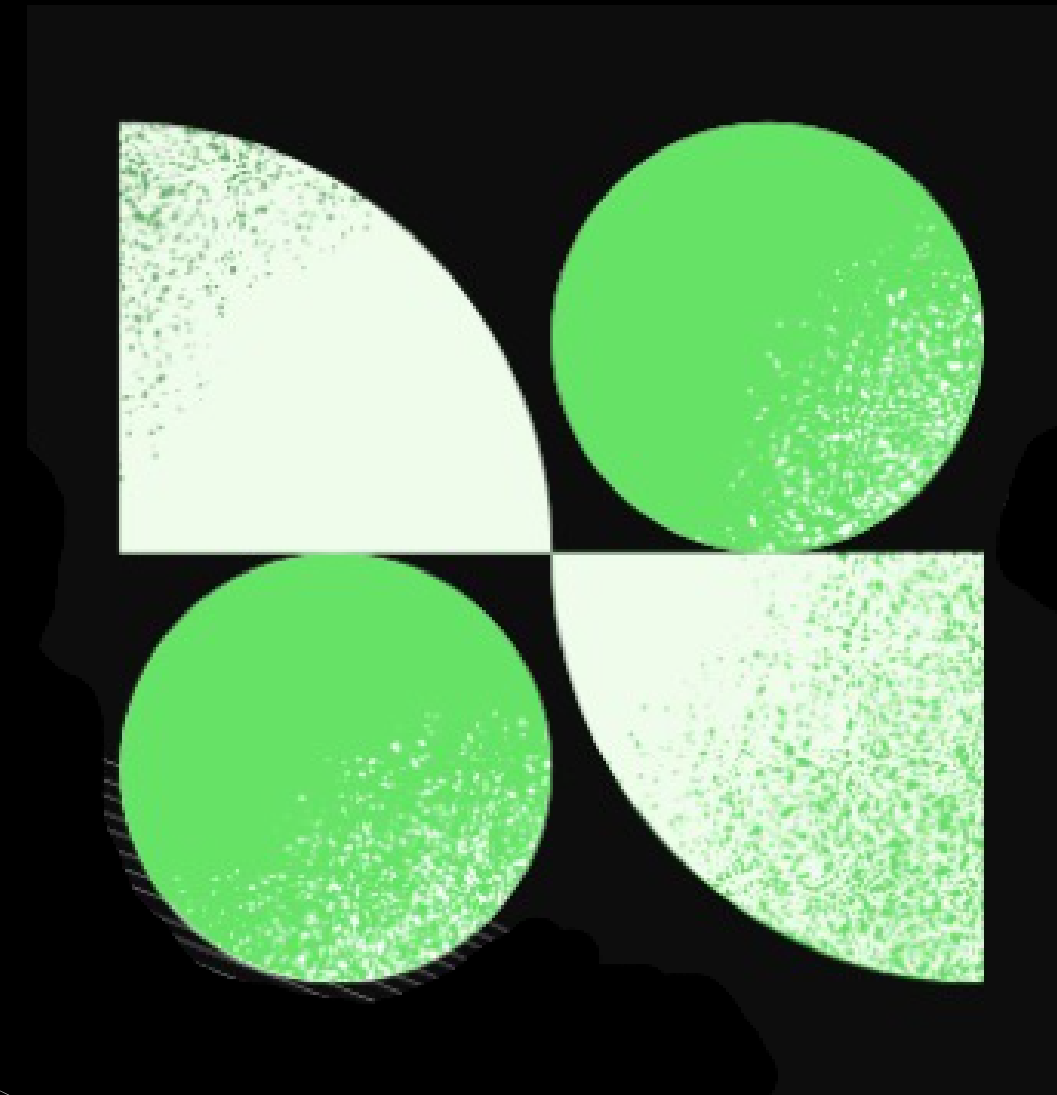


SOCBIZ SUMMER OPEN PROJECT 2026

Shreshtha Shrinivas-24124040



Agentic AI-Based **Dynamic Tariff** **Optimization** for EV Charging Networks



Contact: shreshtha_s@bt.iitr.ac.in

TABLE OF CONTENT

Data Landscape and Preprocessing

Key EDA findings and demand behavior insights

Dynamic Tariff Optimization Logic & Pricing Outcomes

Monitoring Agent Evaluation & Feedback Loop Performance

Business, Operational & Policy Implications

Data Landscape and Preprocessing

DATA SOURCES

- ACN-Data | 30,000+ EV charging sessions (Caltech/JPL)
- UrbanEV | 24,798 charging piles with 5-minute interval records

PREPROCESSING

Cleaning → Alignment → FE

- Standardized timestamp formats
- Removed duplicate and invalid records
- Handled missing values using median imputation
- Aggregated sessions into hourly demand profiles

FEATURE ENGINEERING

- Hour of Day & Day of Week
- Peak/Off-Peak Indicator
- Charger Utilization Rate
- Occupancy Density
- Queue Length Proxy
- Revenue per Session

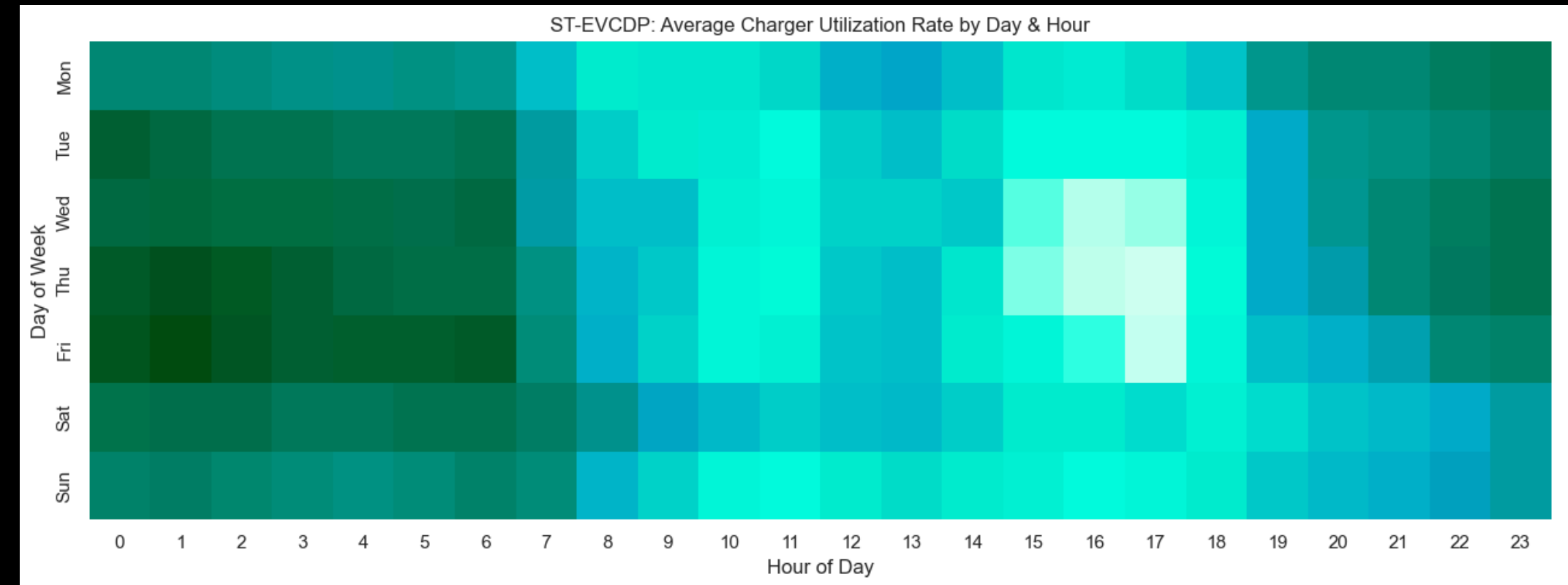
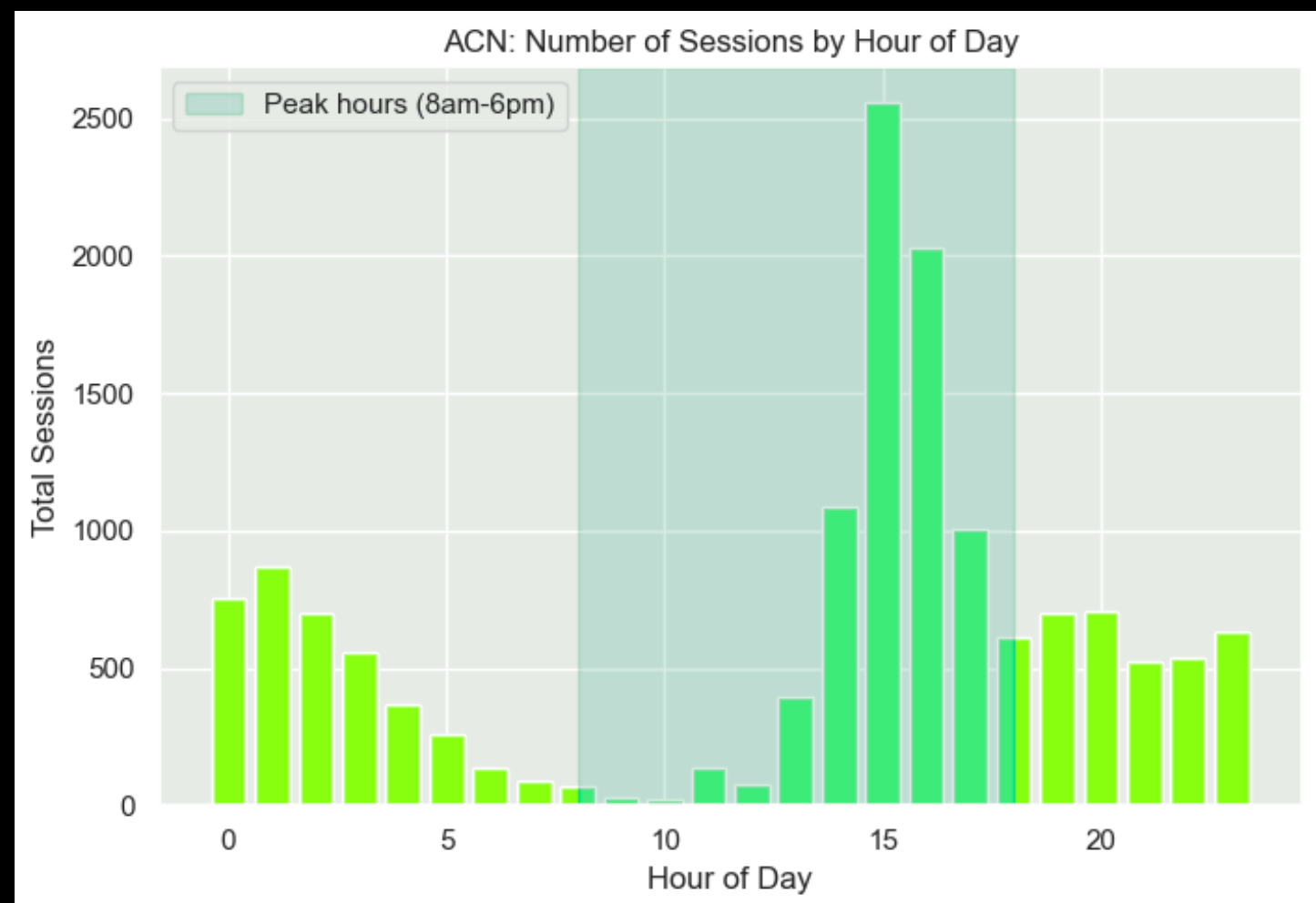


RESULT

Unified, clean dataset enabling demand forecasting, dynamic tariff optimization, and utilization analysis.

Key EDA findings and demand behavior insights

- **Peak charging activity** occurred around **3 PM (15:00 hrs)** in the ACN dataset, indicating clear demand concentration periods.
- **14,848 charging sessions** analyzed, with an average session duration of **5.5 hours** and average energy delivery of **8.94 kWh per session**.



- Chargers experienced **congestion during 66.1%** of observed time intervals, which significantly outnumber **low-demand periods of 16.2%**.
- Significant utilization variability was observed across time periods, highlighting opportunities for congestion management through **dynamic pricing**.

Predictive Performance

- **Random Forest** was found to be best demand prediction agent:
- **For ACN Dataset: ($R^2=0.2863$) For Urban_EV Dataset: ($R^2=0.9892$)**

Dynamic Tariff Optimization Logic & Pricing Outcomes

PRICING FRAMEWORK

• **Baseline Tariff:**
₹15/kWh

• **Dynamic Tariff Rules:**

Utilization	Action
>80	Surge Pricing
30-80%	Standard Pricing
<30%	Discounted Pricing

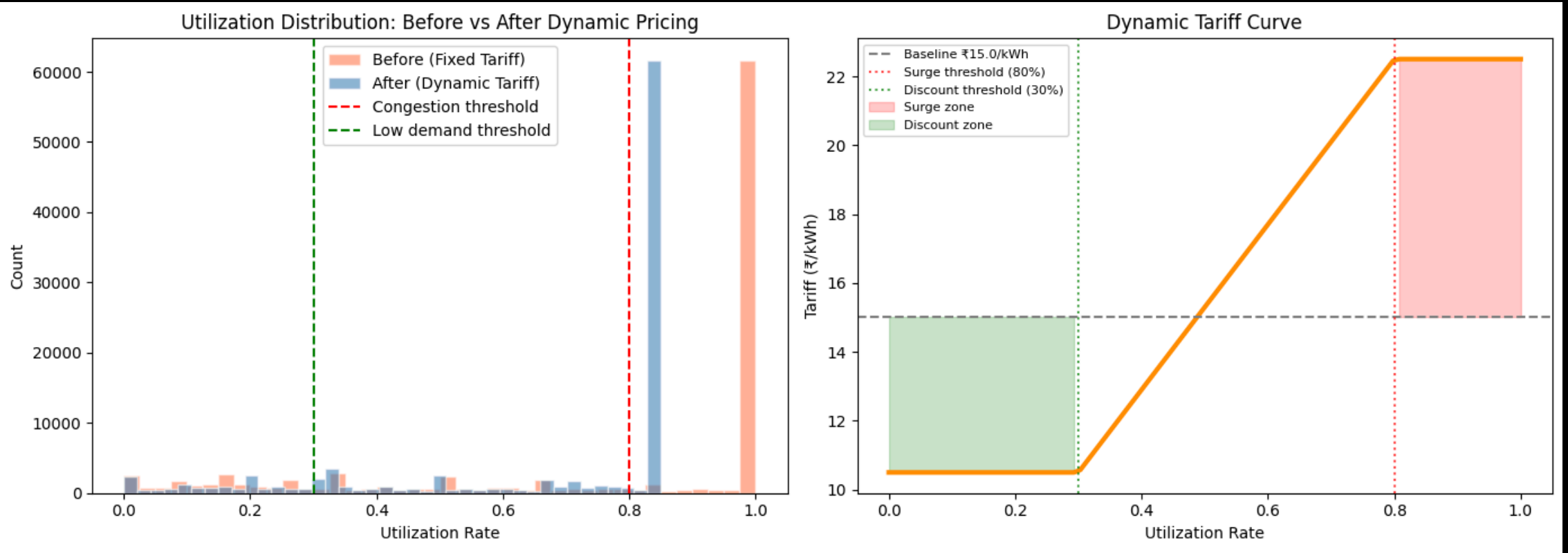
Congestion Impact

- Avg Utilization Before 0.771
 - Avg Utilization After 0.677
 - Congestion % Before 66.3%
 - Congestion % After 63.0%
- **Dynamic pricing also significantly improved reducing the problem of congestion.**

Revenue Impact

Metric	Value
• Baseline Revenue	₹54810967.32
• Dynamic Revenue	₹81070265.9
• Revenue Gain	+47.91%

• **Dynamic pricing significantly improved revenue generation.**



Monitoring Agent Evaluation & Feedback Loop Performance

Evaluation Metrics

Operational Metrics:

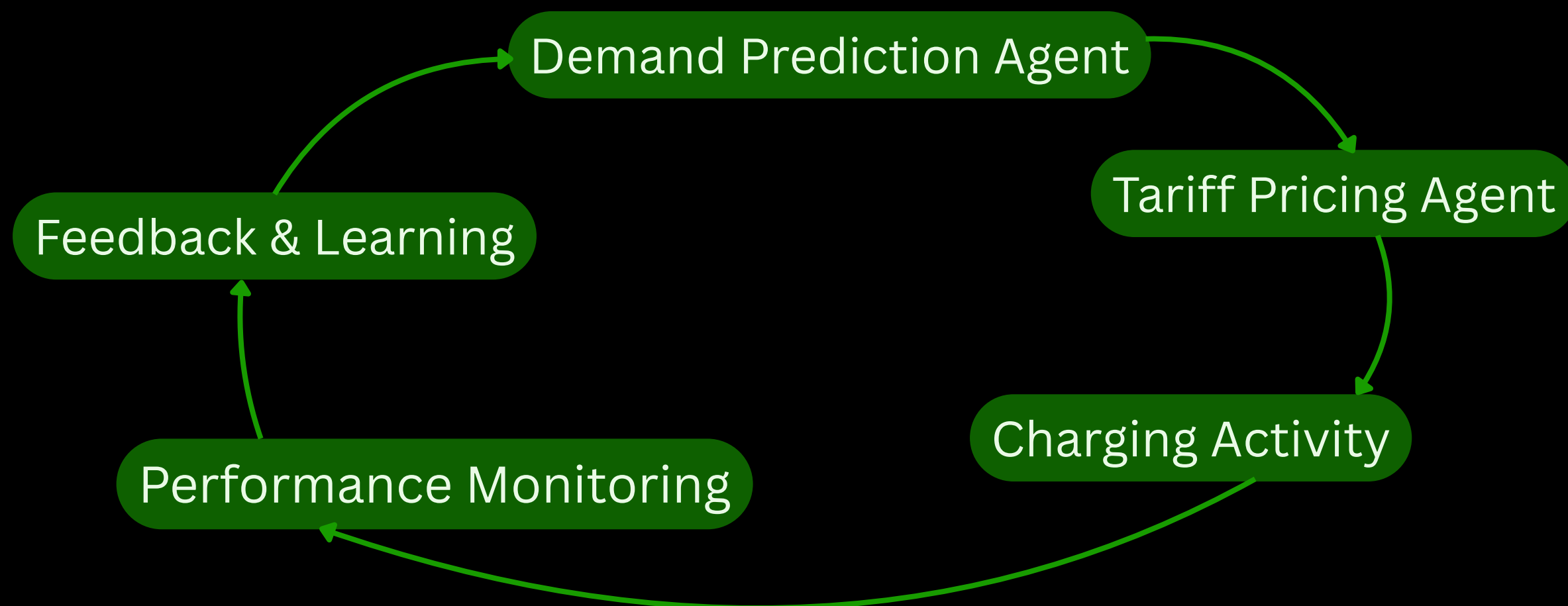
- Avg Wait Time Before 6.382 mins
- Avg Wait Time After 1.558 mins
- Wait Time Reduction 75.6%

Behavioral Metrics:

- Customer Response Rate % 100.0
- Surge Zone Decisions % 66.3
- Discount Zone Decisions % 16.2

Operational Metrics:

- Avg Revenue Gain % across Episodes 47.91
- Avg Pricing Efficiency Score 1211.41
- Prediction RMSE ST-EVCDP 0.0353
- Prediction R2 ST-EVCDP 0.9892



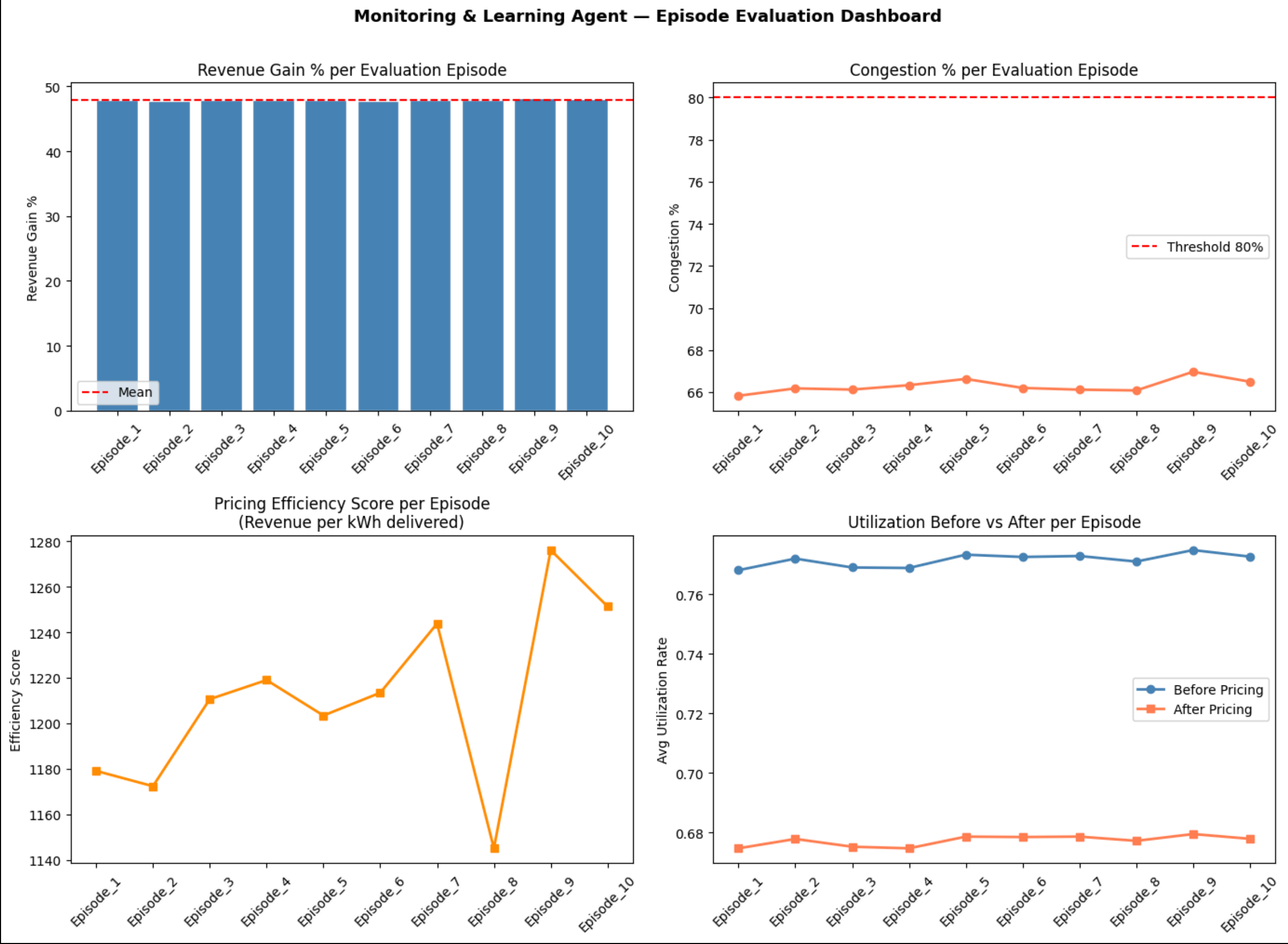
RATING



Result:

The monitoring framework confirms that pricing interventions produced measurable improvements in network performance.

Monitoring Agent Evaluation & Feedback Loop Performance



Business, Operational & Policy Implications

Revenue Optimization

- Increase in revenue
- Better monetization of charging assets
- Reduced inefficiencies from flat-rate pricing

Improved Network Efficiency

- Better charger allocation
- Reduced peak-period pressure
- More balanced station utilization

Customer Experience Impact

- Potential reduction in charging queues
- Improved charger availability
- Incentives for off-peak charging

Dynamic tariffs support:

- Demand-side management
- Load balancing
- Reduced stress on local power infrastructure

Scalability & Future Extensions

- Smart city charging ecosystems
- Fleet charging operators
- Highway fast-charging corridors
- Renewable-energy-aware charging networks

FINAL CONCLUSION

By combining demand forecasting, adaptive pricing, and continuous monitoring, the proposed Agentic AI framework transforms EV charging from a static pricing system into a data-driven, self-improving operational platform that increases revenue, improves utilization, and supports sustainable EV infrastructure growth.

Thank You!

